

Rabin Shrestha

Senior Software Engineer · AI & LLM Applications

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SUMMARY

Software engineer with 12+ years of backend and distributed-systems experience, now building **LLM-backed products end to end**. Builds the infrastructure that makes LLM products viable in production — multi-provider routing through LiteLLM and OpenRouter, per-user usage and cost tracking from day one, and local inference options so sensitive data never leaves the user's machine. Develops spec-first alongside AI coding agents on production codebases. Previously **founding engineer** at ZenLedger — helped grow it from startup stage through **Series B**, led a team of up to 15, and owned the ingestion platform behind **10B+ transaction rows** and **\$50B+ in tracked assets**.

EXPERIENCE

Marpha Labs — Co-Founder & Engineering Lead

Remote · 2025 – Present

An independent product studio: a distributed collective of builders across Nepal, the USA, Canada, and Australia, self-funding and shipping products to test them in market.

FillAnyForms PRE-LAUNCH

LLM infra · Rails 8.1 · AWS

- Built an **AI-assisted document processing system** that extracts fillable fields from PDFs — and from scans via image-to-text — and maps structured user data into them, on a Rails 8.1 API with Sidekiq keeping inference off the request path.
- Built the LLM layer end to end: a **LiteLLM proxy** over OpenRouter giving the platform multi-provider model routing, virtual API keys per consumer, and **per-user usage and cost tracking** persisted server-side — so spend is attributable and capped per account, not discovered on the monthly bill.
- Added a **privacy mode** routing inference to the user's own Ollama or OpenAI-compatible endpoint, so document data never leaves their machine — one mapper interface, cloud or local backend behind it.
- Built a companion browser extension as a **user-controlled identity wallet** (BIP-39, did:key, client-side encryption), keeping the personal data that feeds the model on the user's own device.

HabreCare IN DEVELOPMENT

React Native · Expo · Supabase

- Building a mental-wellbeing app for anxiety, insomnia, and overthinking on Expo SDK 54 and React Native 0.81 in strict TypeScript, for iOS and Android.
- Develop **spec-first with AI coding agents**: written design specs and a decision log per feature, committed agent instruction files, and automated typecheck, lint, and test gates every agent-authored change must pass before merge.
- Architected **local-first sync** — writes land on-device instantly and offline, then reconcile to Supabase under per-user row-level security — with media uploading through short-lived presigned Cloudflare R2 URLs minted by edge functions.

SwiftScout LIVE ON APP STORE

Native iOS · Swift

- Shipped a native iOS app alerting drivers to **51 enforcement cameras across San Francisco and Oakland**, processed **entirely on-device** for privacy; on the App Store since April 2026 across six releases.
- Took the app from first commit through App Store review and pricing as a paid subscription product, after learning Swift and the native iOS toolchain from a Rails/Go background.

ZenLedger.io — Founding Engineer & Lead Software Engineer

Remote · 2018 – 2025

Crypto tax and accounting platform. Joined at startup stage as a founding engineer and scaled the product and engineering team through its Series B round, on a team distributed across North America, Europe, and Nepal.

- Owned the data ingestion and parsing platform integrating **100+ cryptocurrency exchanges and blockchains**, processing **10B+ transaction rows** and tracking **\$50B+ in customer assets** across **100K+ customers**.
- Led an engineering team of **up to 15**, setting engineering standards, owning delivery timelines, and driving architecture and specification reviews across teams.
- Served as the core engineer on the tax calculation and automated report generation pipeline, and **rewrote the calculation service from Rails/Sidekiq to Go — 70% less compute** for the same workload, running **60% faster**, cutting compute spend proportionally.
- Built and operated the payment system, and designed **autoscaling AWS infrastructure** that matched capacity to highly seasonal demand, protecting availability at peak while holding down off-peak cost.
- Built services in Ruby on Rails, Go, and React, with background processing on Sidekiq and Faktory, instrumented through Grafana, Datadog, and ScoutAPM.

EB Pearls — Software Engineer → Associate Project Manager

Kathmandu · 2013 – 2018

- Delivered **6+ enterprise projects** for an Australia-based client in PHP (CodeIgniter, Laravel) and MySQL, from requirements through release.
- Promoted into delivery ownership as Associate Project Manager — **ran delivery across** engineering, QA, and design, owning scope changes, budgets, and release timelines.

TECHNICAL SKILLS

AI & LLM — LLM application design · LiteLLM · OpenRouter · multi-provider routing · virtual keys, usage & per-user cost tracking · local-inference privacy modes (Ollama, OpenAI-compatible) · image-to-text extraction · spec-driven development with AI coding agents

Exploring — LLM internals · transformer architectures

Core — Go · Ruby on Rails · TypeScript · React · PostgreSQL · Redis · AWS · Docker

Strong — React Native (Expo) · Next.js · Swift / native iOS · Supabase · Sidekiq · Faktory · OpenTofu/Terraform · Cloudflare (R2, CDN) · GitHub Actions · RSpec · Jest · Playwright

Familiar — MySQL · PHP (Laravel, CodeIgniter) · Preact · Grafana · Datadog · ScoutAPM · CircleCI · Maestro

Leadership — Team leadership (up to 15 engineers) · architecture & delivery ownership · Agile/Scrum · Certified ScrumMaster (CSM)

EDUCATION

B.E. Computer Engineering — Kathmandu University, Dhulikhel, Nepal

2009 – 2013

Certified ScrumMaster (CSM) — Scrum Alliance